

Achievr

AI Accountability Companion

Final Updated Project Proposal / Report

Project Title: Achievr – AI Accountability Companion

Institution: CSIE Department

Course: 3rd Year Final Project

Developers: Brightwel, Olwethu, Thotshiswa

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2. Executive Summary

Achievr is an AI-powered goal setting and accountability system designed for students and working individuals who struggle with consistency, planning, and motivation. Unlike traditional to-do apps, Achievr provides **AI-generated sub-goals, adaptive smart reminders, location-based accountability** via verification, and **weekly performance reports**, and a **social platform where you can add friends and monitor their consistency**, allowing users to not just set goals—but achieve them.

Built using **Flutter** for cross-platform development and **Python Flask/Django** for backend intelligence, Achievr integrates custom NLP processing, behaviour analytics, and GPS-triggered reminders to give users personalized, data-driven support throughout their week.

Achievr solves major real-world problems: vague goal setting, lack of routine, poor planning, and insufficient feedback causing people to fall back into unproductivity. The system empowers users with structured, measurable sub-goals and adaptive motivation.

This proposal outlines the motivation, problem context, system design, implementation plan, architecture, database models, and sequence flow.

3. Motivation

Many students and professionals understand *what* they want but rarely know *how* to get there. They set broad goals (“be better”, “get fit”, “study more”), but fail because:

- The goal has no structure.
- They cannot measure progress.
- They lose consistency after a few days.
- They lack real accountability.
- No system assesses their habits or gives weekly feedback.

Achievr exists to solve these issues.

3.1 End-to-End Example: Why Achievr Matters

Example Goal:

"I want to improve my academic performance this semester."

This is vague. The user does not know:

- What tasks to start with
- How to break the goal into manageable steps
- How often to study
- How to track progress
- How to maintain consistency

AI Breakdown Example

Using NLP + templates, Achievr generates structured sub-goals:

Study Habits

- Attend all lectures Mon–Fri
- Study 1 hour after each lecture
- Join a study group for challenging subjects

Assignments

- Start each assignment within 48 hours of release
- Break into - research → outline → draft → revise

Health

- Sleep at least 7 hours daily
- Take a 10-minute break every 90 minutes

Tracking

- Log quiz results
- Review weak subjects every Friday

The user selects or edits these suggestions, which will have required plotting into a weekly schedule. How? It will generate these subgoals into cards that can be plotted into a timetable and depending on how long the task life is, will remain on there until the goal is discontinued.

During the Week

- **Location-based check-ins:** When the user arrives at the library → "Ready to study?"
- **Smart reminders** adjust based on missed or completed tasks.

- **App tracks productivity patterns** (time of day, location, consistency).

At the End of the Week

Achievr generates a **personalized AI report**:

- Total hours studied
- Strongest days
- Missed commitments
- Recommendations for next week
- Streak achievements

It then readily posts it for your friends/advisor/trainer to be able to see. This is something no standard to-do app provides:

Structure → Accountability → Feedback → Motivation.

4. Problem Statement

Most people set goals but fail to achieve them due to predictable behavioural and structural issues.

4.1 Concrete Problems Achievr Solves

Problem 1: Vague & Unstructured Goals

Examples:

- “Be healthier”
- “Study more”
- “Work on my business idea”

These goals cannot be measured or tracked.

Achievr Solution: AI-generated sub-goals structured into actionable steps.

Problem 2: Inconsistent Routine & Habit Drop-Off

Students often:

- Miss study days
- Forget about tasks
- Lose motivation after week 1: which is a global problem amongst all age groups

Achievr Solution: Smart adaptive alarms and verification mechanisms based on behaviour patterns. Notification to accountability partners (Friends, trainer etc.), and even a deduction to your investment Jar (Money you will optionally be allowed to put into the app. It will be deducted slowly into the system as a last resort to incomplete tasks)

Problem 3: Lack of Real Accountability

People plan to go to the gym or library but do not commit.

Achievr Solution: Location-based accountability reminders. Location verification will be one of a few options of verification for an outdoor task. During the first three task execution it will prompt for your location which it will then imbed as the location of that task (if it occurs at a gym far from your home location)

Problem 4: No Feedback Loop

Most apps do not tell users:

- What they did well
- What habits work for them
- How they can improve

Achievr Solution: Weekly AI performance reports with insights and encouragement.

5. Proposed Solution

Achievr provides:

- AI-driven goal structuring
- Progress tracking
- Location verification
- Smart reminders
- Weekly analytics
- Motivational gamification via social groupings

The system acts as a digital coach and a strict accountability partner

6. Core Features

6.1 AI Goal Breakdown

- Uses NLP and pre-designed templates
- Identifies verbs, tasks, durations, and categories
- Generates clear, measurable sub-goals
- Users can edit or delete suggestions

6.2 AI Weekly Reports

- Behavioral analytics (completed tasks, streaks, best productivity times)
- Personalized suggestions
- Motivation summaries
- Performance metrics and productive hours chart

6.3 Smart Alarm Reminders

- Uses local notifications + Firebase
- Adaptive frequency
- Behavior-based adjustment
- Encouraging messages

6.4 Location-Based Accountability

- Geolocation triggers:
 - “You’re at the gym—start your workout?”
 - “Library entered—study for 30 minutes?”
- Automatically logs check-ins

6.5 Gamification Dashboard

- Streak counters
- Badges
- Progress charts
- Weekly score system

6.6 Social Accountability

- Share progress with friends
- Accountability partners

7. Technical Overview

Layer	Tech	Description
Frontend	Flutter (Dart)	Cross-platform mobile UI
Backend	Flask / Django	REST APIs, AI integration, data logic
Database	PostgreSQL / Firebase	Storage for goals, subgoals, reports, locations
AI	spaCy, NLTK, Pandas, Jinja2	NLP, behavior analytics, weekly text generation
Notifications	Firebase Cloud Messaging	Remote reminders
Scheduling	Cron / Cloud Scheduler	Weekly report automation
Location	Flutter Geolocator	GPS detection

8. AI Customization Details

8.1 NLP Goal Analysis

Customized templates:

- Academic goals
- Fitness goals
- Health goals
- Productivity goals

AI extracts:

- Intent
- Action verbs
- Time durations
- Frequency
- Context (study, fitness, personal)

This is not generic AI — it is **custom-tuned rule-based NLP**.

8.2 Behavioural Analytics Engine

- Tracks user activity
- Detects trends
- Produces personalized suggestions

Custom logic example:

- If user misses 3 tasks → reduce workload suggestion
- If user has strong productivity on Wednesdays → propose weekly focus block

8.3 Weekly Report Text Generation

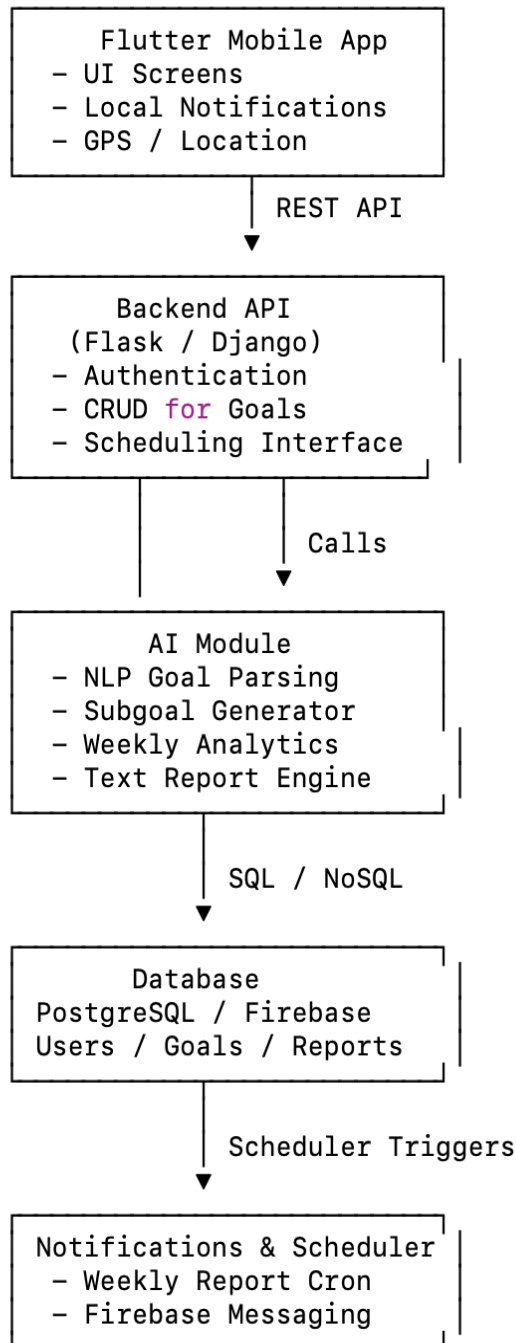
Uses **Jinja2 templating** to ensure safe, clean, academic-level output.

9. System Diagrams

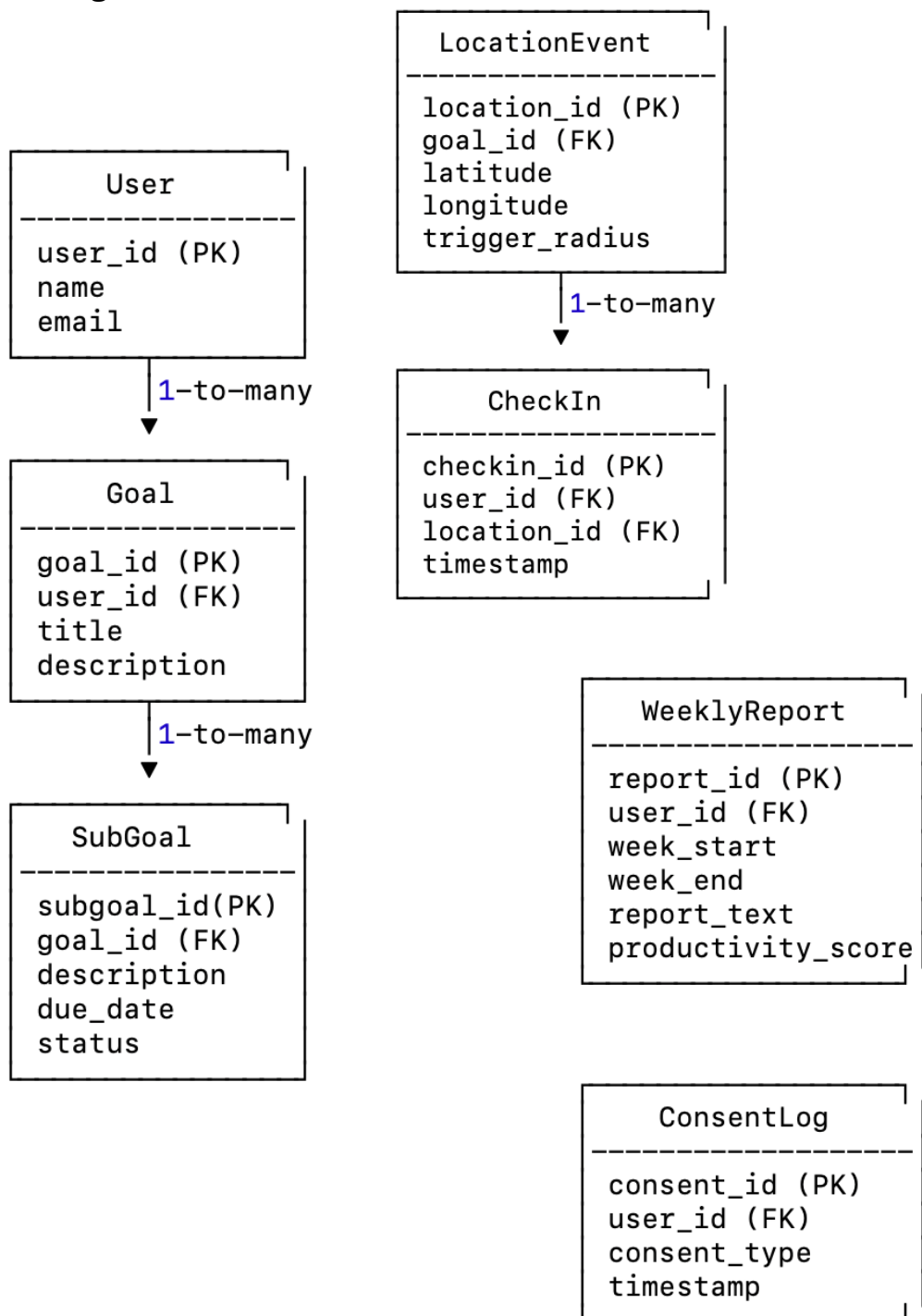
9.1 Architecture Diagram

Explains:

- Flutter → Backend API → AI Engine → DB → Scheduler → Back to App

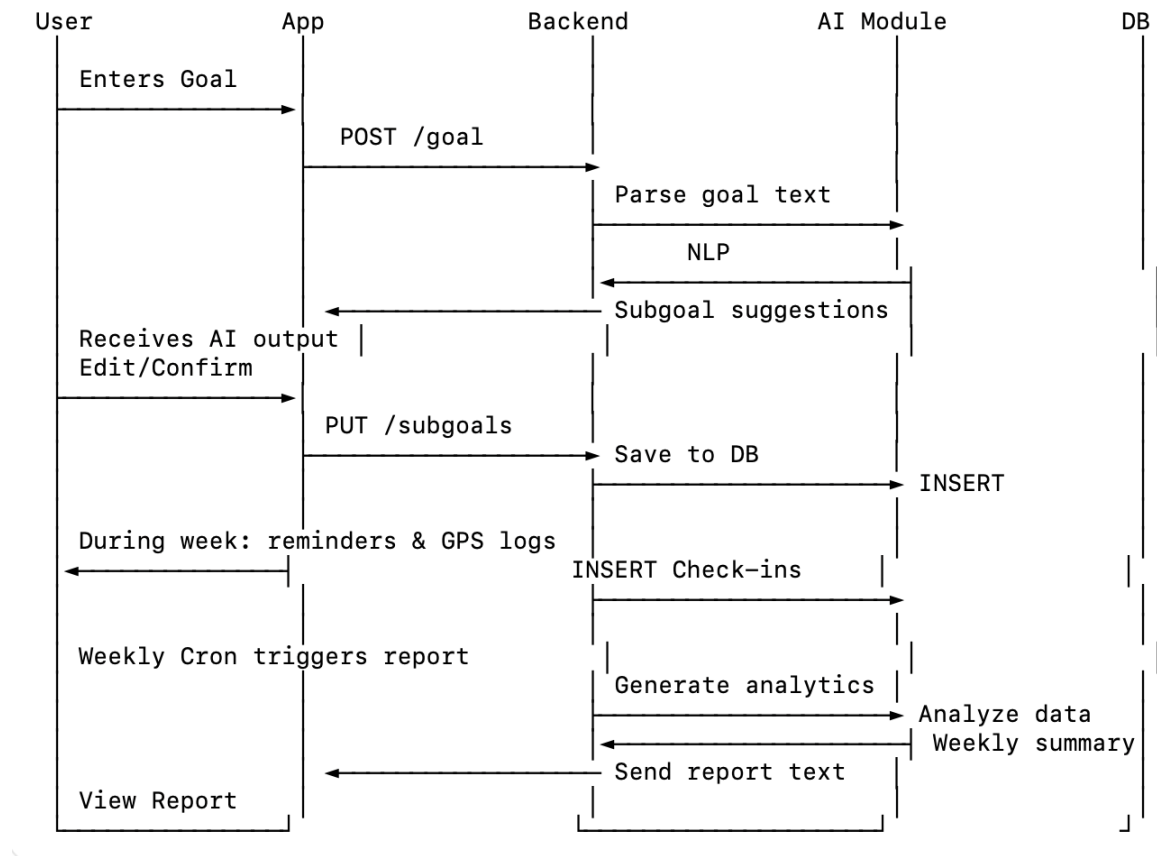


9.2 ER Diagram



9.3 Sequence Diagram

Add goal → process with AI → save → weekly analytics → report generation → display



10. Implementation Plan

Phase	Duration	Tasks	Deliverables
Planning	2 weeks	Requirements, diagrams, roles	Approved proposal
Design	3 weeks	Figma UI, ERD, architecture	Design pack
Development	8 weeks	Backend + AI + Flutter	Working prototype
Testing	3 weeks	Debugging, location tests	Stable build
Deployment	1 week	Firebase hosting, APK build	Final project

11. Expected Outcomes

- Functional AI-powered productivity app
- Real-time accountability features
- Weekly automated reports
- Demonstrated improvement in user consistency
- Strong technical integration of AI + GPS + mobile UI

12. Conclusion

Achievr is a uniquely structured accountability system combining AI, behavioural analytics, scheduling, and geolocation. It addresses real student productivity challenges through structured goal generation, adaptive reminders, and personalized weekly insights. Achievr stands as an impactful project that integrates AI with real-world human behavioural support.

13. References

- Fabulous App
- Habitica
- Coach.me
- Flutter Official Docs
- Flask/Django Docs
- spaCy NLP